

Table 3: Summary of notation.

Variable	Meaning
n	Number of news articles
$\mathcal{A}$	Set of news articles
E	Set of named entities
S	Set of stories
X	Topic distribution vectors in $[0, 1]^m$
D	Viewpoint matrix
C	Viewpoint collection of subsets of A
Q	Subsets of selected items from A
$Q^{\leq j}$	Subset of items in Q assigned to position j or earlier
τ	Target distribution
$\pi(Q, w)$	Weighted average distribution by Q and w
G_{H^2}	Overlap measure based on Hellinger distance
k	Budget parameter for algorithms
t	Number of articles per story

A DETAILS OF THE ANNOTATION PIPELINE

In this section, we introduce the **Diverse News Annotation (DNA)** pipeline for processing news articles. At a high level, the pipeline utilizes semantic annotations to enhance news datasets by incorporating best practices such as news story discovery, topic modeling, entity extraction, and stance detection. Our DNA pipeline code and datasets are available online [4].

Additional notation. We introduce some additional notation that will be useful for explaining the implementation of our pipeline and we also summarize our notation from the main text in Table 3.

For each news story $S_j \in \mathcal{S}$, we denote its articles by writing $S_j = \{a_1^{(j)}, a_2^{(j)}, \dots, a_{|S_j|}^{(j)}\}$. We assume articles in each story are in chronological order and each story has a certain duration $[t(a_1^{(j)}), t(a_{|S_j|}^{(j)})]$. Furthermore, we let $\mathcal{A}^t \subset \mathcal{A}$ denote the subset of n_t articles published before time t . We let $S_j^t = S_j \cap \mathcal{A}^t$ contain all articles from story S_j published until time t ; we set $\mathcal{S}^t = \{S_j^t : S_j^t \neq \emptyset\}$ to all stories that arise until time t . Note that the size of the story set $|\mathcal{S}^t|$ is a monotonic non-decreasing function of t .

Topic modeling. We approach topic modeling as a soft-clustering problem, leveraging BERTopic [13] and its best practices to generate article embeddings, clusters, topic representations, and topic distribution vectors, denoted as $\mathbf{x}_i$. Below, we provide a brief overview of BERTopic and describe our parameter settings for topic modeling.

Method overview. BERTopic is a topic modeling framework that combines transformer-based embeddings with class-based TF-IDF (c-TF-IDF) to produce interpretable topics. It operates through a modular pipeline, which includes embedding, dimensionality reduction, clustering, tokenization, weighting, and optional fine-tuning. Each step is independent, making BERTopic highly customizable.

More concretely, The embedding module converts articles to numerical representations (vectors with fixed length). The dimensionality reduction module reduces these high-dimensional vectors to lower-dimensional ones, as clustering methods struggle with high-dimensional data. The clustering module identifies clusters of various shapes. The tokenization module generates the bag-of-words representation of each cluster (by merging articles in the same cluster) and then computes the class-based TF-IDF score to

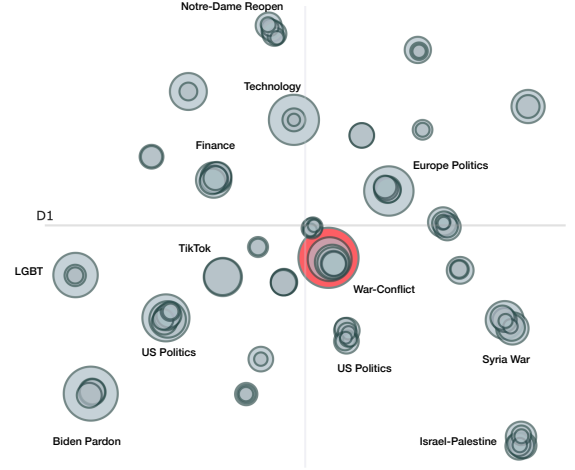


Figure 6: Visualization of topic modeling result on Google-News dataset. Bertopic applies the LDAvis technique to visualize the clustered articles.

compute topic representation (important words per topic according to c-TF-IDF). Finally, given the set of keywords and representative articles in each topic, fine-tuning modules is optional to improve the topic quality by using advanced methods. Each step in the process is independent of each other. Therefore, BERTopic is modular and allows us to customize topic models through a variety of sub-model choices in each step.

Parameter setting. We follow the sequencing process outlined in BERTopic and customize the topic modeling based on the best practices reported by BERTopic. Specifically, we adopt sentence-transformer [29] "all-MiniLM-L6-v2" to compute article embeddings. The embeddings are passed through a UMAP [22] module to reduce their dimension to 5. We adopt the density-based clustering method HDBSCAN [7] to find clusters of different shapes and set the minimum cluster size as 15. In the tokenization module, we limit the top representative words per topic to 10. For fine-tuning, we use the Maximal Marginal Relevance (MMR) method with a diversity parameter of 0.2 to minimize duplicate words in each topic. Finally, we compute topic distribution vectors $\mathbf{x}_i$ for each article using the topic distribution approximation method, with a window size of 4 and a stride of 1.

Figure 6 illustrates the topic clusters identified on the Google-News dataset. Key topics extracted include significant themes such as the Syrian War, the Israel-Palestine conflict, and others.

News story discovery. We approach the news story discovery by hard clustering methods, i.e., each news article is assigned to only one story. Here, we employ the method (*USTORY*) proposed by Yoon et al. [37] to compute the stories S_j . This method achieves the best performance in our experiments, compared to other baselines [20, 23, 33] as well as their variants, while it is robust and scalable. We briefly summarize the process of *USTORY* and how we set the parameters for news discovery.

Method overview. At a high level, the method considers the set of articles $\mathcal{A}$ as a time-based stream and it clusters the articles based

on their similarity, as computed in a sliding time window of a given size. It follows a sequence of steps including temporal theme identification, theme/time-aware embedding and novel story discovery. More concretely, *USTORY* models temporal themes by retrieving keywords through a tokenization pipeline that generates a diverse set of relevant keywords. Given a temporal theme (keywords), it computes the embedding of each article a_i by pooling sentence representations, weighted by their relevance to the theme, which is determined by the frequency and importance of keywords in each sentence. The method maintains a time window $\hat{t}$ and initializes stories $\mathcal{S}^{\hat{t}}$ with thematic summaries based on articles from the first window. As the window moves forward with a fixed stride and reaches t_i , the model processes each new article $a_i \in \mathcal{A}^{t_i} \setminus \mathcal{A}^{t_i-1}$, comparing its similarity to the thematic representations of existing stories. The article is then either assigned to an existing story or initiates a new one. The module also updates the thematic representations of existing stories based on the current time window.

Parameter setting. In practice, we set the window size and slide size of the sliding time window to 7 days and 1 day, respectively. The COVID-Jan and NELA-Jan contain 31 windows (one month), while WCEP-2018 contains 365 windows (one year). We set the minimum number of articles to form a story to be 30. For sentence embeddings, we use Sentence-BERT [29]⁴. We adopt the TF-IDF method as the scoring function, set the number of thematic keywords to 10, and use a temperature of 2 for scaling the confidence scores.

Entity extraction. The entity extraction module extracts the target entity set E of the article set $\mathcal{A}$ by taking into account background information about politicians, political parties, and countries. The method follows 3 steps to obtain the target entity set E including raw entity extraction, entity label refinement and target entity selection.

Raw entity extraction. First, we extract article entities using the BERT-based named entity recognition (BERT-NER)⁵ on a sentence level. It returns the raw entity set $\tilde{E}_i$ of article $a_i \in \mathcal{A}$ with pre-defined entity types fall into {“Person”, “Location”, “Organization”, “Miscellaneous”}.

Entity label refinement. The entity types provided by BERT-NER are general and lack specificity for downstream tasks. To enhance data quality, we postprocess these results by matching entities with Wikidata. For instance, the entity “Donald Trump”, labeled as “Person” by BERT-NER, is refined to “Politician” based on Wikidata. Due to the efficiency issues when accessing Wikidata SPARQL API⁶, we constructed two offline datasets POLITICIAN and PARTY. The POLITICIAN dataset contains metadata of 294,323 politicians born after 1940, including their name, country, sex and date of birth. The PARTY dataset includes the names and abbreviations of 2,198 political parties, along with their affiliated countries. In this paper, we only consider politicians, political parties and counties as target entity types. The post-processing is automated by matching entities of type “Person” with the POLITICIAN dataset, “Organization” with the PARTY dataset, and “Location” identified as country names, to refine entity set.

Heuristic for target entity selection. We observe that the entities contained in each article are in large numbers, and the complexity of the articles makes it challenging to identify meaningful target entities for a given article. However, based on our assumption in the news story discovery, each story contains correlated articles with a unique theme. A small subset of entities may be *theme-aware* entities and appear in most of the articles within a story. For example, news articles in the story “2020 United States Presidential Debate” should contain the same politician entities *Donald Trump* and (or) *Joe Biden*. The two entities are both highly related to the story theme, and it is meaningful to use these two entities as the target for stance detection.

Inspired by the observation, we introduce a heuristic to filter a meaningful subset of target entities in each article at the story level. More concretely, we denote $\Omega = \{\text{“Politician”, “Political Party”, “Country”}\}$ to be the target entity type set. For each article $a_i \in \mathcal{A}$, we write $E_{i|\Omega}$ to be the candidate entities of a_i with entity types in Ω . Similarly, for each story $\mathcal{S}_j \in \mathcal{S}$, we write $E_{\mathcal{S}_j|\Omega} = \{\cup_{a_i \in \mathcal{S}_j} E_{i|\Omega}\}$ to denote the set of entities with entity type in Ω of story $\mathcal{S}_j$. Then the frequency of each entity $e_k \in E_{\mathcal{S}_j|\Omega}$ is defined as $f(e_k) = \frac{\sum_{a_i \in \mathcal{S}_j} \mathbb{I}(e_k \in E_{i|\Omega})}{|\mathcal{S}_j|}$ where $\mathbb{I}(\cdot)$ is an indicator function that equals 1 if $e_k \in E_{i|\Omega}$ and 0 otherwise. Intuitively, $f(e_k) \in [0, 1]$ indicates the ratio articles that contain entity e_i . And $f(e_k) = 1$ indicates that entity e_k appears in every article in story $\mathcal{S}_j$. We introduce a threshold parameter $\gamma \in [0, 1]$ and select entities with $f(e_i) \geq \gamma$ as target entities. We write $E_{\mathcal{S}_j|\Omega, \gamma}$ to denote the resulting subset of entities in type Ω in story $\mathcal{S}_j$. Intuitively, this heuristic chooses entities in meaningful types that appear in at least γ -fraction of the articles in a story. In practice, we set $\gamma > 0.2$ and obtain the target entity set $E_{\mathcal{S}_j|\Omega, \gamma}$ for each story $\mathcal{S}_j$ and the target entity set $E = \cup_j E_{\mathcal{S}_j|\Omega, \gamma}$.

Stance detection. We consider stance detection as a prediction task on a predefined category. Specifically, we aim to predict the stance of a given news article towards a given entity (*target*) within the categories {“in-favor”, “neutral-or-unclear”, “against”}. This helps us track stance bias in news articles. The stance detection module first utilizes the entity extraction module to obtain entity set E . Then it adopts the zero-shot learning approach to obtain stance detection results. In practice, we adopt the FLAN-T5 model [9] for stance detection. More concretely, we take each article a_i and each target entity of a_i to create a prompt following the best practice in zero-shot learning. Figure 7 show the structure of the prompt used for stance detection task.

Note that our goal is not to improve or invent stance detection methods. Instead, we consider the state-of-the-art stance detection models as a black box. The stance detection module could be replaced when better methods come out. We use a zero-shot learning approach and employ FLAN-T5 model [9] for stance detection.

New source bias. We obtain the political leaning of news sources from Media Bias/Fact Check (MBFC) [1], as it is a widely used resource for evaluating bias and reliability of news sources [26, 32]. We deploy an open-source data collection tool⁷ to collect the media source bias annotation from MBFC. The media source bias rating

⁴<https://huggingface.co/sentence-transformers/all-roberta-large-v1>

⁵<https://huggingface.co/dslim/bert-base-NER>

⁶<https://query.wikidata.org/>

⁷https://github.com/CASOS-IDeaS-CMU/media_bias_fact_check_data_collection

Given a news article, what is the stance of the news article below with respect to '{target entity}'? If we can infer from the news article that the news supports '{target}', please label it as 'in-favor'. If we can infer from the news article that the news is against '{target}', please label it as 'against'. If we can infer from the news article that the news has a neutral stance towards '{target}', please label it as 'neutral-or-unclear'. If there is no clue in the news article to reveal the stance of the news article towards '{target}', please also label it as 'neutral-or-unclear'. Please use exactly one word from the following 3 categories to label it: 'in-favor', 'against', 'neutral-or-unclear'. Here is the news article. '{article text}' The stance of the news article is:_____.



Figure 7: Prompt structure for stance detection of each article.

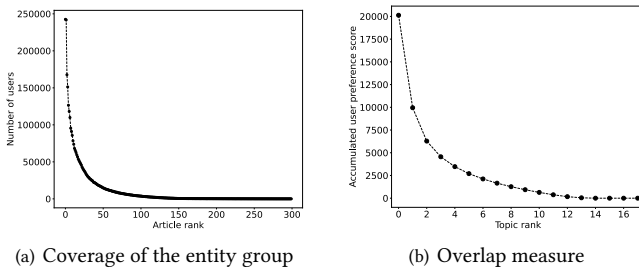


Figure 8: User behavior from Microsoft news dataset.

falls into 5 categories [“left”, “left-center”, “center”, “right-center”, “right”] where “center” is considered to be least biased on *MBFC*, and we label each article with the leaning of its news source.

News consumption behavior of users. Due to the lack of real user consumption data in our dataset, we generate synthetic user behavior data that matches the statistical properties of real user consumption behavior observed in Microsoft News [35].

User click behavior. First, as shown in Figure 8(a), we observe that when presented with a list of recommended news articles, the click rate decreases exponentially as the rank increases. This observation inspires us to simulate user news consumption behavior by selecting the top- k articles from a list. We adopt this as a subroutine for TopNews to model user browsing behavior.

User click behavior. Next, we analyze the user preference distribution across all possible topics. Figure 8(b) shows that user preferences are highly skewed: most users are only interested in a few topics (with high scores). Based on this observation, we simulate user preference distributions with exponentially decaying properties.

ETHICAL CONSIDERATIONS

In developing algorithms aimed at diversifying news consumption, we acknowledge the inherent duality of technological applications: the same tools designed to broaden perspectives can, paradoxically, be repurposed to narrow them. This duality is not merely theoretical but a realistic risk, evidenced by the manipulation of social media algorithms to foster echo chambers rather than dismantle them. Despite these risks, our work underscores the critical importance of addressing echo chambers, thereby exerting pressure

on platforms to continue refining their technologies to support a healthier information ecosystem.

Our operational definition of diversity, focused primarily on political viewpoint diversity, admittedly adopts a narrow scope. This approach is intentional, designed to simplify the complex landscape of news perspectives into a more manageable framework, specifically within the U.S. context characterized by the prevalent left-right divide. This choice facilitates the development and implementation of our methodologies but may not capture the full spectrum of diversity that exists within and across societies. We recognize the limitations of this approach and encourage future research to explore and implement alternative definitions that capture other dimensions of diversity. Moreover, our U.S.-centric approach aims to constrain the problem space to enhance the feasibility of our solutions. However, this geographical and cultural limitation may not directly translate to or be effective in other regions with different political and media landscapes. Acknowledging these limitations is crucial as it frames the applicability and scalability of our findings.

Ultimately, our research seeks to contribute to a broader discourse on the necessity of diverse news consumption. By demonstrating the feasibility of coding and implementing political viewpoint diversity, we hope to inspire the adoption and rigorous real-world testing of these techniques. Such endeavors will be vital for evaluating the efficacy of our methods and understanding their broader implications. Through this work, we aspire to foster an information environment where users are empowered with the choice to access a multiplicity of perspectives, thereby enriching their understanding and engagement with the world.